Minimum Specifications

Title: Prototype GM Tubes (two-year Blanket Purchase Agreements (BPAs))

1. Scope

Naval Surface Warfare Center Carderock Division (NSWCCD) Radiation Detection Technology Branch (RDTB), through Naval Sea Systems Command (NAVSEA) 04RD, has been tasked with procuring prototype Geiger-Mueller (GM) tubes as an alternative to sustain fielding of the Multi-Function RADIAC (MFR) survey meter used throughout the Naval Nuclear Propulsion Program.

2. Background

GM tubes are needed for use in the MFR, specifically the control unit (IM-265) and the gamma-beta probe (DT-680). The MFR has been in use for over 20 years with over 10,000 units in the field. The current GM tube has become more unreliable with increased failure rate. Thus, the RDTB aims to find an alternative, accurate and durable GM tube to sustain the MFR.

3. Objective and Product Specification:

The NSWCCD RDTB intends to establish multiple-award blanket purchase agreements (BPAs) for the purpose of ordering prototype GM tubes, per paras. 3.1 and 3.2, below.

This specification document applies to each of the contractors/manufacturers listed in Enclosure (1) Manufacturers List.

3.1 (BPA Year 1) Upon establishment of the BPA an initial order is anticipated of up to three (3) GM tubes with the following characteristics and specifications:

Characteristic ↓	**Specification**
Sensitivity*** 137Cs cpm at 1 mR/h	270
Recommended Operating Voltage	460
Plateau Length Volts (min.)	420-520
Plateau Slope (%/100 V max.)	30
Dead Time (µs max.)	60
Background (cpm)*** Shielding 2” Pb + 1/8” Al	25 max.
Operating Temp (°C)	-20 to +60
Cathode Material	Cr/Fe
Cathode Wall	280 mg/cm^2

*** At recommended operating voltage

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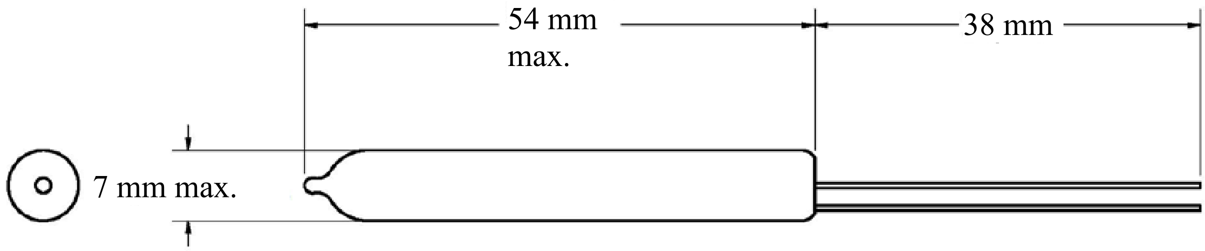

- Exposure rate range: 0.01 mR/h – 1,000 R/h
- Thickness of lead wires pictured above: 0.5 mm maximum diameter
- Navy will perform testing of each GM tube with ARCADES (Automated RADIAC Calibration and Diagnostic Equipment System) software at 70 R/h and 30 mR/h to ensure attainment of a Sensitivity Factor (SF), calculated within the software, in the 5.0 to 7.0 range. (The Navy uses ARCADES to acceptance test GM tubes for field use. The calculated SF is a check to confirm that the GM tube performs at the Sensitivity referenced above.)

3.2 (BPA Year 2) Following Navy testing, in the event a subsequent BPA order is issued, the contractor shall provide a quantity of up to five (5) GM tubes according to the same para. 3.1, above, characteristics and specifications.

4. Place of Performance
Material shall be packaged at Contractor facility.
Material is Free on Board (FOB) Destination:
Naval Surface Warfare Center Carderock Division
Radiation Detection Technology Branch
9500 MacArthur Boulevard
West Bethesda, MD 20817
Attn: Mark DiNezza, B157

5. Period of Performance
BPA Year 1 – Initial Order: Quantity of up to three(3) GM tubes
BPA Year 2 – Final: Quantity of up to five(5) GM tubes shall be delivered 90 days after any placed order that may occur between 180 days and 365 days after date of the established BPA.

6. Point of Contact
TBD